

PRODUCT REQUIREMENTS DOCUMENT

VayuAI

AI-Powered Multi-Source Daily Weather Intelligence for India

Version 1.0 | June 2026 | CONFIDENTIAL

Document Information

Product Name	VayuAI — AI-Powered Weather Intelligence Platform
Version	1.0 (Initial Release)
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Target Market	India — all 28 states and 8 UTs
Document Owner	Product Team
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1. Executive Summary

VayuAI is an AI-powered, mobile-first daily weather intelligence platform purpose-built for India. It aggregates real-time and forecast data from multiple free weather APIs, fuses them using a proprietary AI ensemble model, and delivers hyper-accurate, day-by-day weather insights — including temperature, rainfall, humidity, UV index, and severe weather alerts — across 5,000+ Indian cities, towns, and tehsils.

India's weather is uniquely complex: six distinct climate zones, violent monsoon swings, dense fog belts, and microclimates that can shift within a 20-km radius. No single free API covers this complexity accurately. VayuAI solves this by intelligently blending data from five or more free sources, filling gaps with AI-generated estimates, and presenting the result in plain, regional-language-ready daily summaries.

2. Problem Statement

2.1 Current Pain Points

- Single-source weather apps produce unreliable forecasts for tier-2 and tier-3 Indian cities.
- Free APIs (Open-Meteo, wttr.in, etc.) each excel at different parameters — no one source is consistently best.
- Farmers, logistics operators, event planners, and daily commuters in India need daily-granular data (not just hourly snapshots).
- Rainfall prediction accuracy in India is notoriously poor in consumer apps during monsoon, pre-monsoon, and cyclone seasons.
- Existing apps do not aggregate or cross-validate sources, leaving users dependent on a single flawed model.

2.2 Opportunity

- India has 700M+ smartphone users, with weather being among the top-5 most searched topics daily.
- Multiple high-quality free-tier weather APIs exist, but none are combined intelligently.
- AI ensemble methods (weighted averaging, ML bias correction) can dramatically reduce forecast error when applied to multi-source data.

3. Product Vision & Goals

3.1 Vision Statement

"To become India's most trusted daily weather companion by turning raw multi-source data into clear, actionable, AI-verified forecasts that every Indian — farmer, student, traveller, or city dweller — can rely on."

3.2 Success Metrics (12-Month Targets)

Metric	Target	Measurement Method
Daily Active Users (DAU)	500,000	Firebase Analytics
7-Day Forecast Accuracy (Rain)	>78%	IMD Ground Truth Comparison
Temperature RMSE vs Actuals	<1.5°C	Automated backtesting
Cities Covered	5,000+	Database count
App Store Rating	>4.4 / 5	Google Play & App Store
API Uptime (aggregated)	>99.5%	Uptime monitoring

4. Free Weather Data Sources

VayuAI pulls from the following free-tier APIs. Each source is polled independently and data is tagged with source metadata before entering the AI fusion pipeline.

API / Source	Free Tier Limit	Key Strengths	Parameters Used
Open-Meteo	Unlimited (non-commercial)	High-res global NWP, no key needed	Temp, Rain, Wind, UV, Humidity (7-day)
wtrtr.in	Unlimited	Simple HTTP, India-optimised	Temp, Feels-like, Rain, Cloud cover
7Timer!	Unlimited	Civil + astronomical layers	Temp, Wind, Precip. probability
Open Weather Map (free)	60 calls/min, 1,000/day	Large city DB, PM2.5/AQI data	Temp, Rain, Snow, AQI, Alerts
WeatherAPI.com (free)	1M calls/month	Realtime + 3-day forecast, monsoon flags	Rain mm, Snowfall, Sunrise/Sunset
IMD Open Data (govt)	Public / scrape	Authoritative India rainfall, cyclone	Station rainfall, cyclone alerts, normals

Note: All free-tier API calls are rate-managed by an internal API gateway with Redis-based throttling. IMD data is fetched via authorised scraping / public datasets and refreshed every 6 hours.

5. AI Fusion Engine

5.1 Architecture Overview

The core differentiator of VayuAI is the AI Ensemble Fusion Engine (AEFE). The AEFE processes raw data from all sources and produces a single, confidence-scored daily forecast for each city.

Step 1 — Data Ingestion & Normalisation

- All API responses are parsed and normalised to a common schema: {date, city_id, temp_max, temp_min, rainfall_mm, humidity_pct, wind_kmh, uv_index, condition_code}.
- Missing fields from any source are marked as NaN and handled by the imputation layer.

Step 2 — Weighted Ensemble Model

- Each source is assigned a dynamic weight based on its historical accuracy per city and per season (monsoon, winter, summer, post-monsoon).
- Initial weights are seeded from backtesting against IMD historical actuals (2019–2024).
- Weights are retrained monthly using new ground-truth data.
- Final value for each parameter = $\Sigma (\text{source_value} \times \text{source_weight}) / \Sigma \text{weights}$ (only non-NaN sources).

Step 3 — ML Bias Correction

- A lightweight XGBoost model per city corrects systematic bias (e.g., Open-Meteo consistently underestimates monsoon rainfall in Kerala by ~8%).
- Correction models are trained on a rolling 90-day window of forecast vs. actuals.

Step 4 — LLM-Generated Daily Summary

- The fused numeric forecast is passed to an LLM (Claude API / Gemini API) via a structured prompt.
- Output: a 2–3 sentence plain-language daily summary per city.
- Example: "Wednesday will be mostly cloudy in Bhopal with a high of 34°C. Expect heavy rainfall in the afternoon (18mm likely), so carry an umbrella if stepping out. Humidity stays high at 82%."
- Summaries are available in English, Hindi, Bengali, Tamil, Telugu, Kannada, and Marathi at launch.

Step 5 — Confidence Scoring

- Each daily forecast is assigned a Confidence Score (0–100) based on inter-source agreement, data freshness, and seasonal variance.
- Score < 50: displayed with a warning icon — "Low confidence, check back later."

6. Core Product Features

6.1 Daily Weather Dashboard

- 7-day daily forecast cards showing: max/min temperature, rainfall probability, rainfall amount (mm), wind speed, humidity, UV index, sunrise/sunset.
- "Today at a Glance" hero card with AI-generated plain-language summary.
- Source agreement indicator: how many of the 6 sources agree on rain today (e.g., 5/6 sources predict rain).

6.2 Rainfall Intelligence Module

- Daily rainfall forecast in mm with confidence band (e.g., 12–22mm expected).
- Monsoon tracker: daily IMD-calibrated monsoon progression map for June–September.

- Cumulative monthly rainfall vs. normal (sourced from IMD 30-year normals).
- Flood risk flag for low-lying cities (integrated with NDMA risk zones).

6.3 Multi-City & Location Management

- Auto-detect current location via GPS.
- Save up to 10 cities (free tier) / unlimited cities (premium).
- Quick-switch between saved locations.
- Coverage: 5,000+ Indian cities, towns, tehsils, and major villages (lat/lon grid-based fallback for unlisted locations).

6.4 Smart Alerts & Notifications

- Push alerts for: heavy rainfall warnings (>30mm/day), cyclone approach (sourced from IMD), extreme heat (>44°C), cold wave, dense fog, and thunderstorm.
- Customisable alert thresholds per user.
- "Will it rain tomorrow?" daily briefing notification (opt-in, 7 AM IST).

6.5 Agriculture & Outdoor Mode

- Farmer-specific daily view: sowing advisories, irrigation recommendations, pest/disease risk (based on humidity + temp combos).
- "Good day to dry crops / dry clothes" binary indicator.
- Outdoor event planner: enter a date + location, get AI-generated risk summary.

6.6 Air Quality Index (AQI) Integration

- Real-time AQI pulled from OpenWeatherMap + CPCB open data.
- Daily AQI forecast (next 3 days).
- Health advisory based on AQI level.

7. Technical Architecture & Stack

7.1 Backend

Layer	Technology	Purpose
API Gateway	FastAPI (Python)	Rate-limit & route all weather API calls
Data Pipeline	Apache Airflow / Celery Beat	Scheduled polling of all 6 sources

Layer	Technology	Purpose
Cache Layer	Redis	Store latest API responses, rate-limit counters
Database	PostgreSQL (TimescaleDB ext.)	Time-series forecast storage & actuals
AI/ML Engine	Python + XGBoost + scikit-learn	Ensemble weighting & bias correction
LLM Summaries	Claude API (claude-haiku)	Daily plain-language summaries
Notifications	Firebase Cloud Messaging	Push alerts across Android + iOS
Hosting	AWS / Railway.app	Scalable deployment, India-region CDN

7.2 Frontend / Mobile

Layer	Technology	Purpose
Mobile App	React Native (Expo)	iOS + Android from single codebase
Web App	Next.js 15	PWA + SEO-friendly city weather pages
State Management	Zustand + React Query	Client-side caching & API sync
Maps	React Native Maps / Mapbox	India weather heatmap layer
Charts	Victory Native / Recharts	7-day trend charts, rainfall bars

8. Data Refresh & Pipeline Schedule

Data Type	Source	Refresh Interval	Priority
Current conditions	OWM + WeatherAPI	Every 30 min	Critical
7-day daily forecast	Open-Meteo + wttr.in	Every 6 hours	High
AI-fused daily output	AEFE pipeline	Every 6 hours post-fetch	High
LLM daily summaries	Claude API	Once per day (6 AM IST)	Medium
IMD rainfall normals	IMD Open Data	Monthly	Low
ML model retrain	TimescaleDB actuals	Monthly	Low

Data Type	Source	Refresh Interval	Priority
Cyclone / severe alert	IMD + NDMA	Every 15 min (active events)	Critical

9. Build Roadmap — Phased Plan

Phase 1 — Foundation (Months 1–2)

- Set up FastAPI backend with API gateway for all 6 weather sources.
- Build Redis caching layer and PostgreSQL/TimescaleDB schema.
- Implement basic weighted ensemble (equal weights initially).
- Build React Native MVP app: location search, 7-day cards, basic alerts.
- Deploy on Railway.app with CI/CD (GitHub Actions).

Phase 2 — AI Core (Months 3–4)

- Integrate XGBoost bias correction models (seed with backtested IMD data).
- Build Celery Beat pipeline for scheduled 6-hourly refresh.
- Integrate Claude API for LLM daily summaries (English + Hindi first).
- Add confidence scoring and source agreement indicator to app UI.
- Launch beta with 100 cities.

Phase 3 — Intelligence & Scale (Months 5–6)

- Expand to 5,000+ cities using lat/lon grid interpolation.
- Add Monsoon Tracker, Flood Risk flag, AQI integration.
- Add 5 more regional languages for LLM summaries.
- Launch Farmer Mode with sowing advisories.
- Firebase push notifications for severe weather alerts.

Phase 4 — Growth & Monetisation (Months 7–12)

- Introduce freemium paywall: unlimited saved cities, 14-day forecast, API access (B2B).
- B2B API product for logistics, agriculture, event companies.
- India weather SEO pages (Next.js) for organic growth.
- ML model retraining pipeline fully automated.
- Target: 500K DAU, 1% conversion to premium.

10. Monetisation Strategy

Tier	Features	Price	Target Segment
Free	7-day forecast, 3 saved cities, ads	₹0	Mass users
Pro (Monthly)	14-day, unlimited cities, no ads, farmer mode	₹49/month	Power users
Pro (Annual)	All Pro features, priority alerts	₹399/year	Loyal users
B2B API	Fused data API access, SLA, custom cities	₹2,999– ₹9,999/month	Enterprises

11. Risks & Mitigations

Risk	Severity	Mitigation
Free API rate limits exhausted at scale	High	Redis caching, staggered refresh, multiple API keys
Free API provider discontinues service	Medium	6-source redundancy; any 3 sources sufficient for ensemble
LLM API costs rise at scale	Medium	Batch generation once/day; cache summaries in DB
IMD scraping blocked	Medium	Use IMD public CSV datasets + POSIX open data feeds
Poor forecast accuracy in new cities	High	Low confidence score warning + user feedback loop
Competitor (Windy, IMD app) improves UX	Low	Differentiate on AI summaries + multi-language support

12. Out of Scope (v1.0)

- International weather coverage (outside India).
- Radar / satellite live imagery (requires paid APIs).
- Hyperlocal (street-level) weather.
- Custom trained LLM (use existing APIs only).
- Apple Watch / wearable app.
- WhatsApp / Telegram weather bot (Phase 5 consideration).

13. Appendix — API Quick Reference

Free API Endpoints Used

- Open-Meteo: https://api.open-meteo.com/v1/forecast?latitude={lat}&longitude={lon}&daily=temperature_2m_max,precipitation_sum&timezone=Asia/Kolkata
- wttr.in: <https://wttr.in/{city}.json?lang=en>
- 7Timer!: <http://www.7timer.info/bin/civillight.php?lon={lon}&lat={lat}&output=json>
- OpenWeatherMap: <https://api.openweathermap.org/data/2.5/forecast/daily> (free key required)
- WeatherAPI: <https://api.weatherapi.com/v1/forecast.json?key={KEY}&q={city}&days=3>

Key India-Specific Considerations

- Always request `timezone=Asia/Kolkata` in all API calls — India is UTC+5:30.
- Monsoon season (June–September): increase source polling frequency to every 3 hours.
- Use IMD District-level rainfall normals CSV as ground truth baseline for model training.
- AQI scale: use CPCB India AQI breakpoints (not US EPA) for health advisories.

End of Document — VayuAI PRD v1.0